

- ❑ Topic of the session: Advanced Java Programming
- ❑ UNIT 10: **HIBERNATE**
- ❑ **University of Kigali (UoK)**
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HIBERNATE



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10.1 Hibernate Framework

Hibernate is a Java framework that simplifies the development of Java application to interact with the database. It is an open source, lightweight, ORM (Object Relational Mapping) tool. Hibernate implements the specifications of JPA (Java Persistence API) for data persistence.

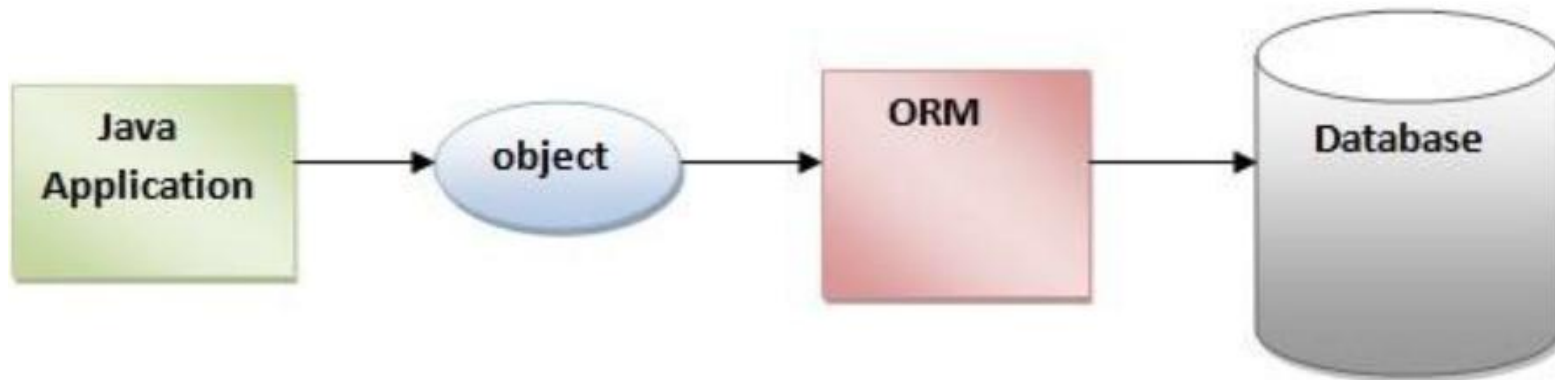
Hibernate is a tool (a framework) in Java that helps you connect your Java program to a database easily.

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10.2 ORM Tool

An ORM (Object Relational Mapping) tool simplifies the data creation, data manipulation and data access. It is a programming technique that maps the object to the data stored in the database.



The ORM tool internally uses the JDBC (Java Database Connectivity) API to interact with the database.

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10.3 Advantages of Hibernate Framework

Following are the advantages of hibernate framework:

1) Open Source and Lightweight (simple, and easy to use.)

Hibernate framework is open source under the LGPL license and lightweight.

2) Fast Performance

The performance of hibernate framework is fast because cache is internally used in hibernate framework. There are two types of cache in hibernate framework first level cache and second level cache. First level cache is enabled by default.

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10.3 Advantages of Hibernate Framework

3) Database Independent Query

HQL (Hibernate Query Language) is the object-oriented version of SQL. It generates the database independent queries. So you don't need to write database specific queries. Before Hibernate, if database is changed for the project, we need to change the SQL query as well that leads to the maintenance problem.

4) Automatic Table Creation

Hibernate framework provides the facility to create the tables of the database automatically. So there is no need to create tables in the database manually.

5) Simplifies Complex Join

Fetching data from multiple tables is easy in hibernate framework.

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10.4 Creating the Database

We use MySQL database called **Sakila**. The sample database is not included when you install the IDE so you need to first create the database.

The Sakila database is a free sample MySQL database that is available from the MySQL site. To create the Sakila database you can download and install the Sakila Sample Database plugin using the Plugins manager. After you install the plugin you can create the Sakila database from the Services window. The Sakila database is added to the list of databases in the Create MySQL database dialog box.

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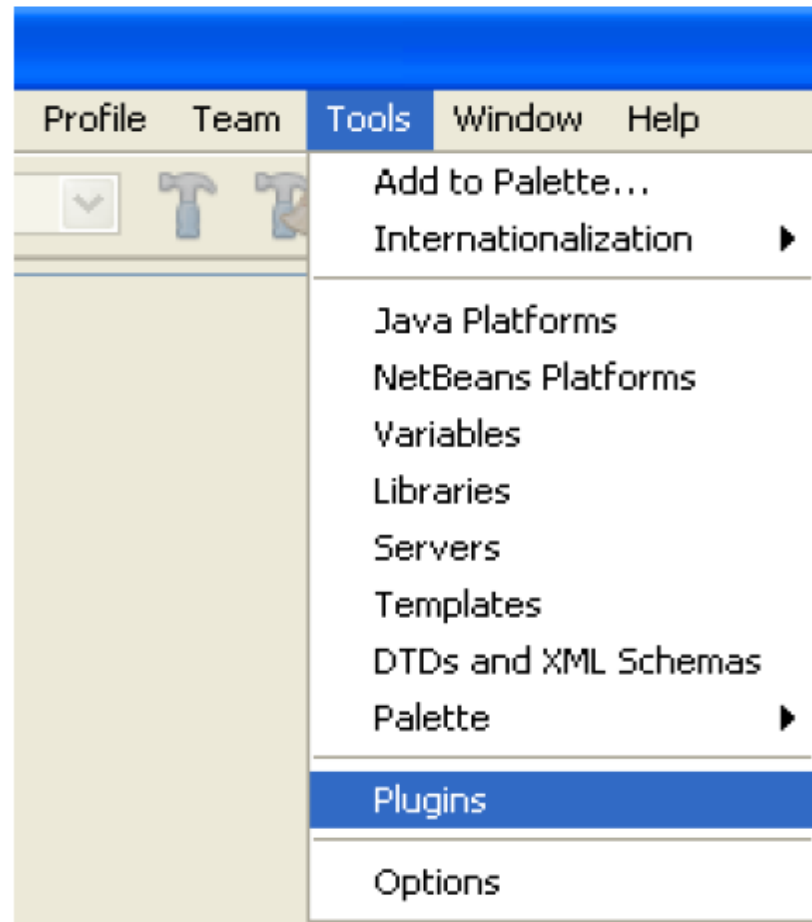
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10.4 Creating the Database

1. **Make sure you are connected to the Internet**
 2. **Open the **Plugins manager (in NetBeans)** and install the Sakila Sample Database plugin.**
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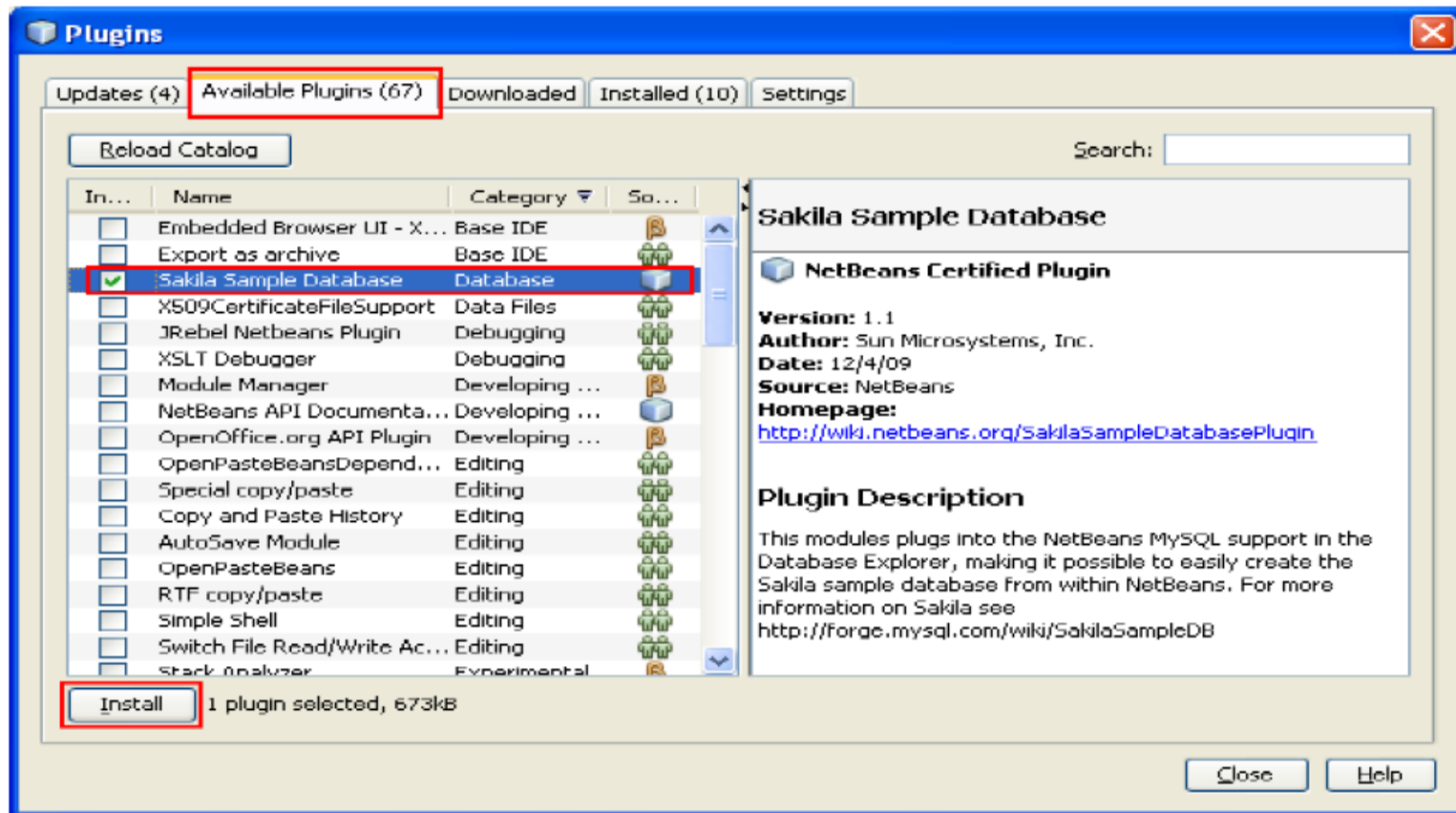
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10.4 Creating the Database



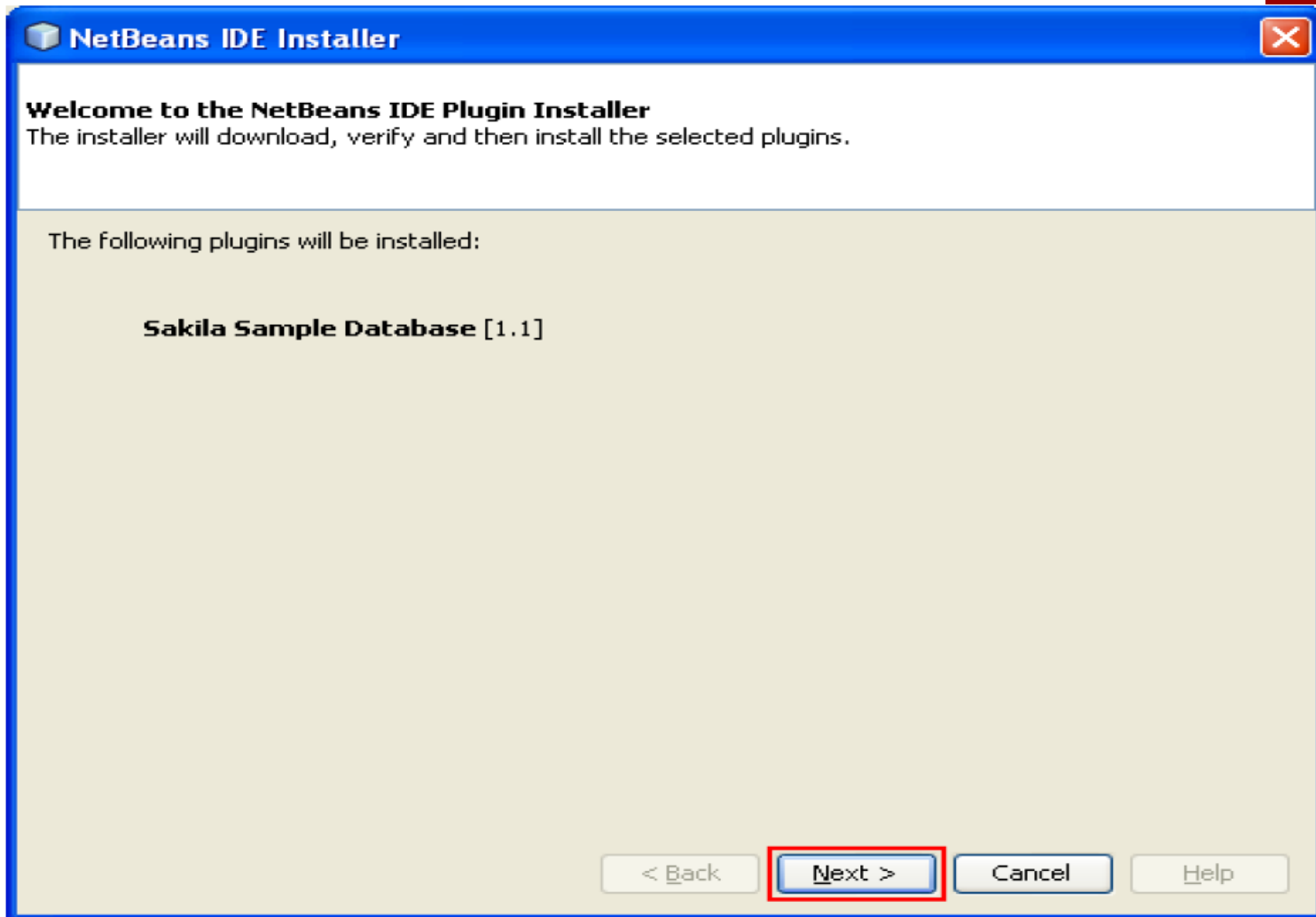
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10.4 Creating the Database



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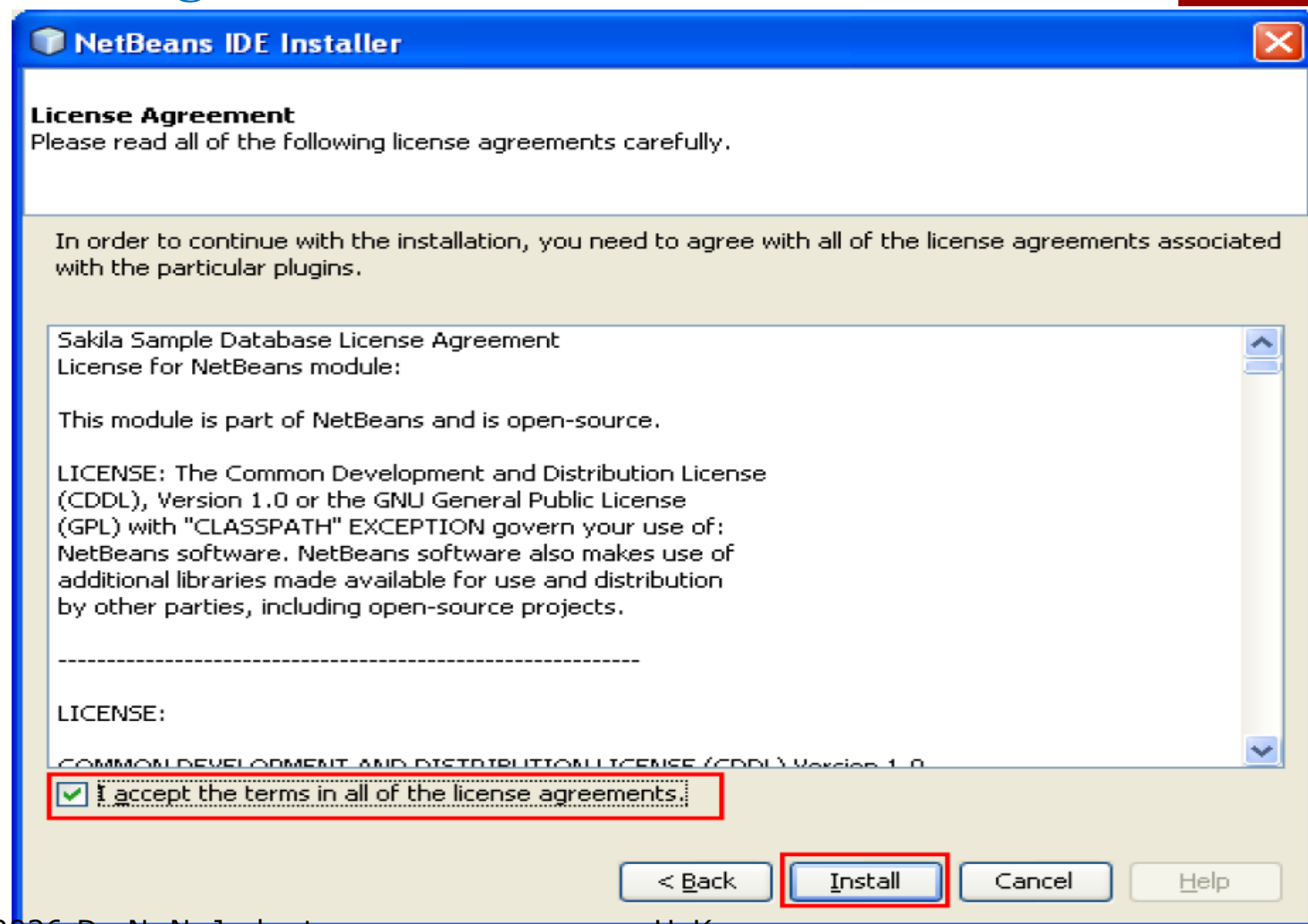
10.4 Creating the Database



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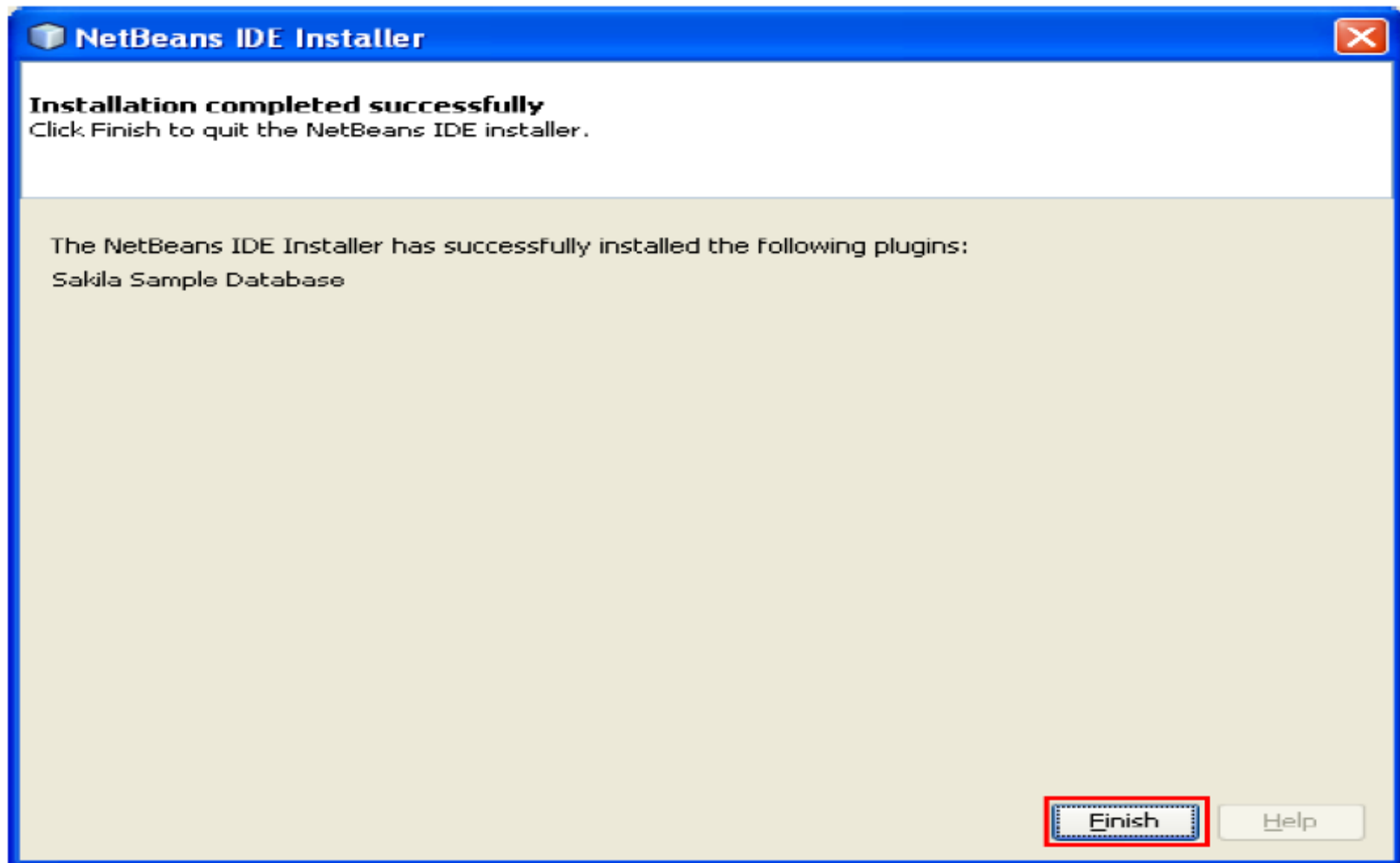
10.4 Creating the Database



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10.4 Creating the Database



10.4 Creating the Database

1. After installing the plugin, start the MySQL database server by expanding the Databases node in the Services window, right-clicking the MySQL Server node and choosing Start.
2. Right-click the MySQL Server node and choose Create Database.
3. **Select the Sakila database from the New Database Name drop down list in the Create MySQL Database dialog box. Click OK.**

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10.4 Creating the Database

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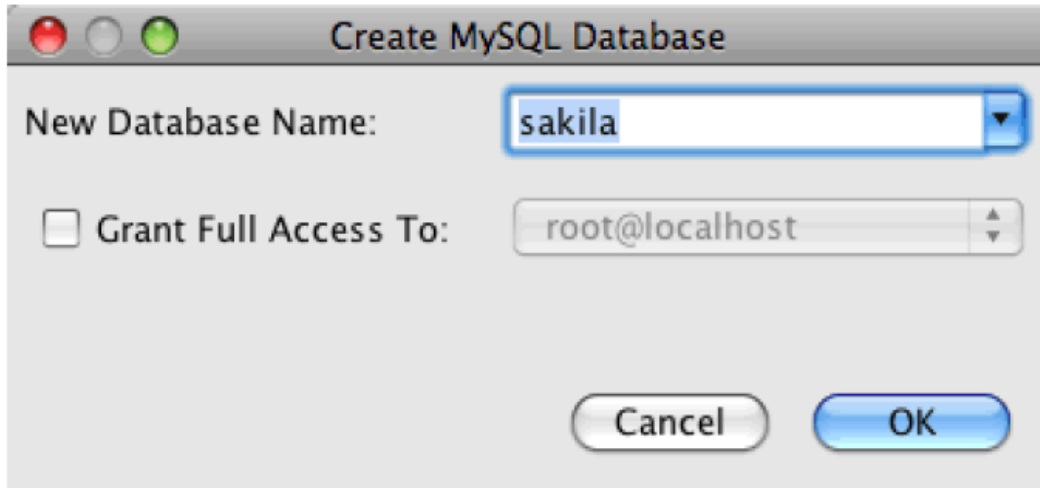


Figure: Screenshot of Create MySQL Database dialog

When you click OK a Sakila node appears under the MySQL Server node.

1.Right-click the Sakila node and choose Connect.

When you click Connect a database connection node for the Sakila database (`jdbc:mysql://localhost:3306/sakila [username on Default]`) is listed under the Databases node.

When a connection is open you can view the data in the database by expanding the connection node.